Carbon Footprint Monitoring System for Smart Campuses Using IoT and Occupancy Analytics

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***Abstract*—University campuses, because they operate continuously,consume a lot of electricity. This includes classrooms, auditoriums,laboratories ,research Centers, data centers ,Server rooms ,dormitories, administrative Offices. This constant energy use of electricity throughout the campus contributes to higher carbon emissions, which highlights the need for smart monitoring and ways to Use that energy more proficiently.**

This paper presents a framework for analyzing campus energy consumption and estimating potential reductions in carbon emissions using threshold-based demand-side management techniques. Historical time-series energy data is used to identify peak consumption intervals and hourly demand trends. A random forest regression model is incorporated to capture temporal consumption patterns and support the prediction of short-term energy demand.

Simulation trials carried out using load reduction threshold levels of 5%, 10%, 15%, and 20% were successful in achieving demonstrable emission reductions. The 20% scenario achieved a reduction of 2948.52 tonnes of CO_2. The proposed framework may be integrated with IoT-based smart meters to enable real-time monitoring, thereby supporting scalable implementation within a university campus.

***Index Terms*—Energy Augmentation, Campus Carbon Footprint,IoT-based Monitoring, Smart Campus, Carbon Emission Reduction, Energy Analytics Data, Sustainable Infrastructure**

I. Introduction

Climate change and the rise in global energy in today's times have made carbon emission reduction an overriding concern across all sectors [8]. In educational institutions, especially on large university campuses, they operate as small-scale urban ecosystems with electricity consumption in bulk due to lighting, HVAC systems, laboratories, computing facilities, and residential hostels [4], [5]. While many campuses aim to become "green" or "sustainable", effective decision-making often lacks detailed data-driven analysis of actual energy usage patterns.

The cutting-edge developments in artificial intelligence (AI) and Internet of Things (IoT) technologies have opened new opportunities for intelligent energy monitoring and thermal efficiency [1], [2], [12]. Smart meters and IoT-based sensors can continuously collect energy consumption data, while AI techniques can deconstruct those patterns, detect dissipation, and suggest maximum yield strategies [14].

However, many institutes still rely on those static energy policies rather than dynamic and data-informed interventions.

In this study, we propose an analytical framework to evaluate campus energy consumption and estimate the potential carbon emission reductions using threshold-driven optimisation. Instead of implementing the costly hardware, the proposed method first focuses on analysing historical time-series energy data to identify the peak usage hours and simulate controlled reduction strategies [9]. By applying different percentage-based thresholds during high-consumption intervals, we estimate achievable emission savings under practical constraints.

The primary contributions of this paper are as follows:

1) A structured data preprocessing as well as time-series analytic framework for campus energy consumption.
2) A threshold-based optimisation simulation model to estimate carbon emission reduction.
3) Quantitative evaluation of emission savings under multiple reduction scenarios.
4) A scalable system that can be integrated with smart monitoring systems enabled by IoT for real-time deployment in the future [2], [5].

The results of this study show how data-driven strategies can significantly reduce carbon emissions without requiring major infrastructure modifications. This study offers a viable guideline for making the change to sustainable and smart campus environments.

II. Proposed Methodology

The proposed framework adopts a data-driven approach to analyze campus energy consumption patterns and estimate potential carbon emission reductions using threshold-based optimization. The methodology integrates time-series analysis with demand-side management (DSM) strategies [9], [14] and

standardized carbon accounting practices [8]. Similar IoT-enabled analytical approaches for improving building energy efficiency have been discussed in recent smart campus and building automation studies [1], [5], [12]. The overall workflow consists of five key stages: data preprocessing, time-series analysis, carbon emission estimation, peak-load identification, and threshold-based reduction simulation.

The experimental evaluation was conducted using a building-level energy consumption dataset comparable to the ASHRAE Great Energy Predictor dataset, which contains time-stamped electricity usage records for multiple institutional buildings. Such datasets are widely used for analyzing institutional energy consumption trends and supporting smart monitoring–based optimization in campus environments [2], [4], [5], [15]. This dataset enabled detailed analysis of consumption trends and assessment of emission reduction potential under different load-management scenarios.

A. Data Collection and Preprocessing

Energy consumption data that is verifiable for the campus' buildings was used as input in the experiments. Time-stamped energy consumption data covering the total number demand for individual building levels were supplied for the campus Preprocessing process involved the following:

- Temporal fields, in both historical and real-time data products, have been standardized to ensure that timestamp fields are in a consistent datetime format, facilitating time series studies [2], [5], [12].
- Generate hourly consumption data to create a temporal consumption trend analysis from the dataset showing peak demand hours [4], [14].
- Handles missing or bad observations (or any other kind of inconsistent entries) to avoid wrong analysis.
- Hourly aggregation for structured time-series modeling: Allows users to organize data into uniform time intervals to better understand peak and normal load patterns [9], [14].

Time-series structuring in order to capture periodic demand variations is a fundamental characteristic of institutional energy systems [1], [5], [14]. Clean and well-organized data is then a precursor to analytical integrity.

B. Time-Series Consumption and Peak Analysis

Data was preprocessed to study hourly consumption trends to determine times of highest demand. This data was then statistically aggregated to calculate the following:

- Mean hourly consumption
- Peak load values
- Daily demand distribution patterns

Peak intervals were defined as hours above a selected threshold, where the threshold was calculated as the total energy use multiplied by the chosen percentile. Targeting high-demand intervals is a common approach in demand-side management (DSM) to mitigate peak loads and improve energy efficiency [9], [14]. Identifying peak intervals at the campus scale affects operational cost and carbon emissions due to increased demand during these periods [4]–[6].

C. Carbon Emission Estimation

In order to assess the environmental impact, the electricity consumption values were converted into equivalent CO_2 emissions using emission factors provided by the Intergovernmental Panel on Climate Change (IPCC) [8]. The carbon emission calculation follows internationally recognized greenhouse gas accounting methodologies [3].

Carbon emissions were calculated as:

$$CO_2 = E \times EF \tag{1}$$

where:

- E represents energy consumption (kWh)
- EF represents the emission factor (kg CO_2 per kWh)

This conversion approach is consistent with the IPCC 2006 Guidelines for National Greenhouse Gas Inventories [8].

D. Threshold-Based Reduction Simulation

To evaluate potential emission reduction without immediate infrastructure modification, a simulation-based threshold optimization model was implemented.

Multiple reduction thresholds (5%, 10%, 15%, and 20%) were selectively applied to identified peak consumption intervals. The adjusted energy demand was computed as:

$$E_{reduced} = E_{peak} \times (1 - T) \tag{2}$$

where:

- T represents the reduction percentage
- E_{peak} represents energy during peak intervals

Threshold-based peak reduction strategies are widely discussed in DSM literature as cost-effective mechanisms for energy optimization [9], [14]. Rather than deploying complex automation systems, this simulation approach evaluates potential savings prior to physical implementation.

The revised total emissions were then recalculated and compared against baseline emissions to determine:

- Absolute emission reduction (tons of CO_2)
- Percentage reduction

E. Evaluation Metrics

The effectiveness of the proposed framework was evaluated using:

- Absolute Carbon Reduction (tons)
- Percentage Emission Reduction (%)
- Peak Load Impact Analysis

These evaluation metrics provide a quantitative foundation for sustainability assessment and are consistent with energy efficiency evaluation approaches in smart building research [1], [4], [15]..

F. Machine Learning-Based Energy Prediction

We have introduced a machine learning–based prediction component to our system to forecast short-term energy consumption on campus to facilitate demand–side management [9]. Historical time-series electricity consumption data were used as the input to the developed predictive models [14].

We employed a Random Forest Regression approach which is well-suited for predicting a continuous target variable and handling multiple inputs. We also extended the inputs with additional temporal features derived from the timestamp, including the hour of the day, the day of the month (to capture weekly/annual periods such as weekends/easter) and seasonal effects [14].

The developed model was used to predict energy demand of buildings and micro-grids for the following days to identify possible future peak-load and demand peaks at early time. The predicted energy demand values are also used to calculate possible future CO_2 emissions based on the emission factor described in Section III-C [8].

The resulting predictive analytics can be used to support future planning for sustainability and to enable the proactive application of demand-side management strategies in smart campus settings [2], [5].

III. System Architecture

The system architecture was created to facilitate an analytical framework for energy consumption monitoring that is based on the Internet of Things. The system architecture was created to facilitate an Internet of Things-based analytical framework for tracking energy usage and calculating campus infrastructure's carbon footprint. The architecture was designed to facilitate future integration with IoT-enabled smart meters and Building Automation Systems (BAS), which are frequently utilized in contemporary smart buildings [1], [2], even if the current implementation mostly makes use of historical energy usage data. The architecture uses a layered design approach, which is common in energy management systems based on the Internet of Things [12].

This structure makes it possible to clearly arrange various system functions while maintaining the system's scalability and manageability. The architecture's machine learning prediction module estimates short-term energy usage by analyzing past demand trends [14]. Administrators can enhance carbon emission monitoring by using these forecasts to identify peak load periods ahead of time. Data Acquisition, Data Processing, Analytics, Emission Estimation, and Decision Support are the optimum primary layers that make up the architecture. From data collection to decision-making support, each layer is in charge of managing a particular phase of the data pipeline [15].

A. Data Acquisition Layer

Data on energy consumption is gathered from various sources by the Data Acquisition Layer. The main input for the system in this study was historical time-stamped campus electricity consumption records. However, this layer can interface with various kinds of sensing devices and monitoring systems in real-world deployments. These could include IoT-enabled smart energy meters, sub-metering systems that track building or zone-level electricity consumption, and occupancy or environmental sensors that offer contextual data on building utilization.

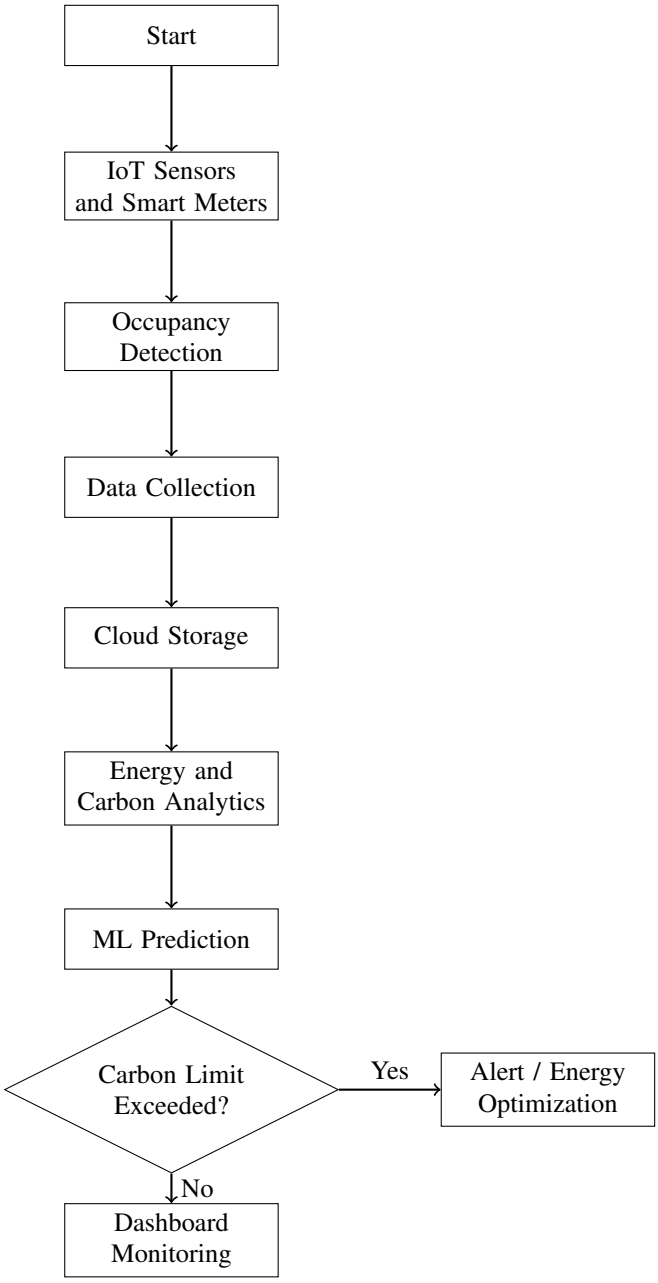

Fig. 1. System architecture of the IoT-based carbon footprint monitoring system for smart campuses

Real-time data collection and IoT-based sensing are widely acknowledged as crucial elements of contemporary smart building energy management systems [2], [3], [5], [12]. Building operators make more precisely analyzed usage trends and

make well-informed judgments about increasing efficiency whenever the energy consumption is continuously monitored.

B. Data Processing Layer

After collection, the raw data is sent to the preprocessing module for additional processing to make the data more reliable and appropriate for analysis for which a number of preparation procedures are used and applied. These procedures involve data cleansing, required to address the missing values, regularizing and formatting time stamps, and aggregating the data into hourly intervals. To study the energy data in a uniform chronological style, time-series structuring is also carried out. This method is crucial since incomplete or inconsistent records are frequently found in real-world energy data causing loopholes in the dataset [2], [5], [12].

This method makes reliability more accurate and reliable input guarantees results for the analytics and machine learning modules by cleaning and organizing the data. Cyclical demand patterns, such as daily consumption peaks or seasonal changes, can simply be revealed by rearranging the data into hourly time-series intervals [4], [14]. Understanding practical behaviour in smart energy systems and guaranteeing accurate downstream analytics always depends on recognizing these patterns [1], [5].

C. Analytics Layer

The analytics layer forms the core processing of the proposed framework. It performs:

- Hourly load profiling
- Peak demand detection
- Statistical trend analysis

Peak load classifying aligns with demand-side management principles which is aimed at reducing energy stress during high-demand intervals [9], [14]. Prior studies also observe that peak reduction analysis plays a crucial role in improving energy efficiency throughout and lowering emissions in institutional buildings and overall campus [4], [6].

D. Carbon Emission Estimation Layer

This layer grabs energy consumption data and translates it into carbon emissions using the IPCC's greenhouse gas accounting rules [8]. So, instead of just tracking energy, the system actually works out the environmental impact on the fly. It goes further than typical energy monitoring by putting those emission estimates right into the main analytics, giving a clearer picture of what's really happening [3], [12].

E. Decision Support Layer

The final layer is all about giving you the insights you need for smart sustainability planning and policy decisions. You get clear baseline emission reports, tools to compare different emission scenarios, and ways to evaluate interventions for peak-load situations. These decision-support tools are key for truly intelligent energy management in smart infrastructure environments [1], [5], [15]. The architecture's modular setup means you can scale up or add new features later without difficulty.

TABLE I
ENERGY CONSUMPTION AND CORRESPONDING CARBON EMISSIONS IN CAMPUS AREAS

Campus Area	Energy Use (kWh)	CO_2 Emission (kg)
Classrooms	1200	540
Laboratories	1850	830
Library	950	420
Administrative Block	760	340

IV. EXPERIMENTAL RESULTS

We tested the threshold-based carbon optimization framework with actual campus electricity usage from past records. To set a baseline, we calculated carbon emissions with standard conversion factors following IPCC guidelines [8]. Next, we picked out the biggest consumption spikes and applied specific reduction thresholds, creating a sort of "what if" scenario to see how demand-side management actions might cut emissions [9]. Four reduction scenarios were simulated: 5%, 10%, 15%, and 20% peak load reduction.

A. Baseline Emission and Peak Load Analysis

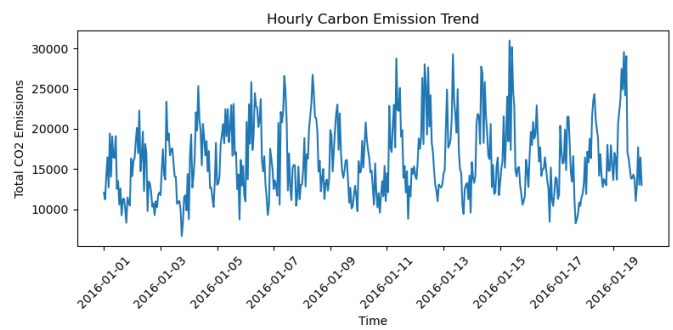

Fig. 2. Hourly Carbon Emission Trend Showing Peak Intervals.

We calculated the baseline carbon footprint by taking the total campus electricity use and converting it into CO_2 emissions, following the approach in Section III-C [8].

When we aggregated the data by hour, clear peak demand periods stood out in the daily load cycle. This kind of peak-heavy pattern is typical for campuses and large institutions, mainly because HVAC systems, lighting, and labs all tend to ramp up at the same times [4]–[6], [14].

Focusing on these high-demand periods fits well with standard DSM strategies designed to ease grid stress and improve operational efficiency [9].

B. Threshold-Based Reduction Results

The simulation results are summarized in Table III. As observed from the results, carbon emission reduction increases with higher peak-load control thresholds. Even a modest intervention of 5% during peak-demand intervals reduces CO_2 emissions by approximately 389.45 tons.

When larger peak reductions are applied, such as a 20% threshold reduction, the avoided emissions increase significantly to nearly 2948.52 tons. The increase in emission savings

is not linear; instead, emission reductions rise sharply as peak-load control becomes more aggressive. This behavior indicates that high-intensity peak intervals contribute disproportionately to overall carbon output, consistent with established demand-side management behavior in institutional energy systems [9].

These findings reinforce observations reported in smart building and smart campus energy optimization studies, which highlight the importance of peak-demand management in reducing institutional carbon footprints [4], [5], [14].

TABLE II
CARBON REDUCTION ACHIEVED USING OCCUPANCY-BASED ENERGY OPTIMIZATION

Occupancy Level (%)	Energy Saving (%)	CO_2 Reduction (kg)
25	8.5	75
50	15.2	140
75	21.7	205
100	26.4	248

TABLE III
CARBON EMISSION REDUCTION UNDER DIFFERENT PEAK REDUCTION THRESHOLDS

Threshold Reduction (%)	Emission Reduction (%)	CO_2 Reduction (tons)
5	0.513	389.45
10	1.543	1172.11
15	2.873	2181.85
20	3.882	2948.52

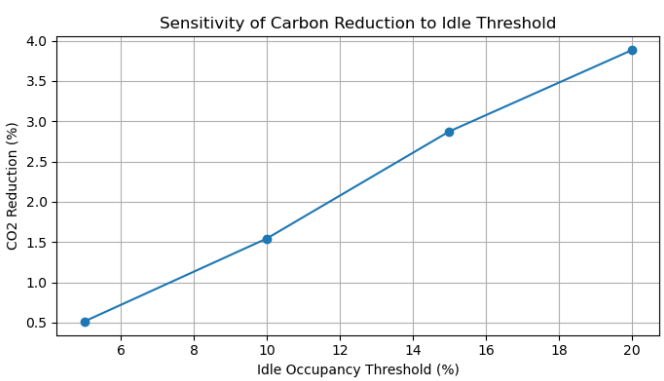

Fig. 3. Carbon Emission Reduction under Different Peak Reduction Thresholds.

C. Demand-Side Management Implications

The confirmed emission reductions demonstrate the effectiveness of targeted peak-load optimization approaches, which are essential components of demand-side management (DSM) architectures [9]. Unlike hardware-intensive automated systems described in earlier IoT-based building automation system (BAS) implementations [1], [13], the proposed simulation-based model provides a low-cost analytical substitute for evaluating sustainability strategies prior to major infrastructure investment.

The findings indicate that:

1) Peak-hour optimization yields disproportionately high environmental benefits.
2) Incremental interventions can achieve measurable sustainability improvements.
3) Analytical modeling can serve as a decision-support precursor to IoT-based automation.

For example, in smart campus studies, the integration of hybrid analytical–IoT frameworks is becoming increasingly important for achieving scalable and economically viable sustainability pathways [2], [5], [15].

V. CONCLUSION AND FUTURE WORK

A. Conclusion

This inquiry confirmed a data-driven methodological framework designed to optimize campus carbon footprints using time-series energy analysis and peak reduction strategies based on threshold principles. The architecture includes preprocessing of clean and structured data, peak load identification, and standardized carbon emission assessment, all consistent with IPCC protocols [8]. Therefore, the proposed model facilitates data-driven evaluation of sustainability measures, removing the requirement for immediate deployment of IoT hardware.

The experimental results show that, together with demand-side management principles [9], [14], targeted peak load reduction can lead to quantifiable carbon savings. A 5% reduction in peak time intervals resulted in avoided CO_2 emissions of 389.45 tons, whereas a 20% reduction produced an approximate emission reduction of 2948.52 tons. These findings confirm that peak demand intervals disproportionately contribute to overall emissions, consistent with observations reported in smart campus energy research [4], [5].

Unlike fully automated IoT-based building control frameworks [1], [12], the proposed technique serves as a scalable decision-support model that allows institutions to simulate sustainability strategies prior to investing in large-scale infrastructure upgrades. The layered architecture also ensures compatibility with IoT-enabled smart meters and real-time monitoring systems for future deployment [2], [15].

Overall, this study highlights the practical role of AI-assisted energy data analysis in advancing campus sustainability objectives and supporting evidence-based environmentally responsible decision-making.

B. Future Work

Although the current framework relies on historical data and simulation-based optimization, several extensions can further enhance its efficiency and impact.

First, integration with real-time IoT-enabled smart meters can enable automated response mechanisms and continuous peak-load tracking in alignment with modern smart building architectures [2], [12]. Incorporating predictive machine learning techniques for short-term load forecasting can further support intelligent demand-side management and proactive energy control strategies [9], [14].

Second, future research can integrate renewable energy sources such as rooftop solar photovoltaic systems to improve supply-demand balancing and reduce dependence on grid-based electricity. IoT-enabled net-zero energy management

approaches can support this transition toward sustainable smart building operation [3], [7].

Finally, extending the proposed framework to district-level or multi-campus deployments would strengthen resilience planning, enable comparative benchmarking, and improve large-scale sustainability assessment. Similar smart-campus monitoring implementations have already demonstrated the effectiveness of IoT-driven optimization in improving institutional energy performance [5], [15].

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